Chapter 5

# Developing Consistent Strategies

IN THIS CHAPTER

- **Obtaining data correctly**
- **Performing automatic data collection**
- **Keeping data relevant**
- **Interacting with data correctly**

Some people view data merely as a disorganized collection of unkempt information from various sources. Throughout this book, you have seen strategies for organizing and manicuring data prior to performing analysis on it. All these strategies see use in business today because data scientists spend an inconceivable amount of time just obtaining, organizing, and manicuring data. Unfortunately, the output of these efforts is often lacking in usefulness because the data simply won't be tamed. The problem is one of consistency in obtaining, organizing, and manicuring the data. If consistency is absent, it's reasonable to assume that the output of an analysis will be suspect. The purpose of this chapter is to define methods of making the act of interacting with raw data more consistent so that the outcome of an analysis is more predictable and, therefore more reliable.

## Standardizing Data Collection Techniques

The data you use comes from a number of sources. The most common data source is from information entered by humans at some point. Even when a system collects shopping-site data automatically, humans initially enter the information. A human clicks various items, adds them to a shopping cart, specifies characteristics (such as size) and quantity, and then checks out. Later, after the sale, the human gives the shopping experience, product, and delivery method a rating and makes comments. In short, every shopping experience becomes a data-collection exercise as well.

REMEMBER You can't sit by each shopper's side and provide instructions on how to enter data consistently. Consequently, the data you receive is inconsistent and nearly unusable at times. By reviewing forms of successful online stores, however, you can see how to provide a virtual self to assist the shopper in making consistent entries. The forms you provide for entering information have a great deal to do with the data you obtain. When a form contains fewer handwritten entries and more check boxes, it tends to provide a better experience for the customer and a more consistent data source for you.

Many data sources today rely on input gathered from human sources. Humans also provide manual input. You call or go into an office somewhere to make an appointment with a professional. A receptionist then gathers information from you that's needed for the appointment. This manually collected data eventually ends up in a dataset somewhere for analysis purposes.

By providing training on proper data entry techniques, you can improve the consistency of input that the receptionist provides. In addition, you're unlikely to have just one receptionist providing input, so training can also help the entire group of receptionists provide consistent input despite individual differences in perspective. Some forms of regulated data entry of this sort have become so complex today that the people doing it actually require a formal education, such as medical data entry personnel (see the course at `https://study.com/articles/Medical_Data_Entry_Training_Programs_and_Courses.html`). The point is that the industry, as a whole, is generally moving toward trained data entry people, so your organization should make use of this trend to improve the consistency of the data you receive.

Data is also collected from sensors, and these sensors can take almost any form. For example, many organizations base physical data collection, such as the number of people viewing an object in a window, on cell-phone detection. Facial recognition software could potentially detect repeat customers.

However, sensors can create datasets from almost anything. The weather service relies on datasets created by sensors that monitor environmental conditions such as rain, temperature, humidity, cloud cover, and so on. Robotic monitoring systems help correct small flaws in robotic operation by constantly analyzing data collected by monitoring sensors. A sensor, combined with a small AI application, could tell you when your dinner is cooked to perfection tonight. The sensor collects data, but the AI application uses rules to help define when the food is properly cooked.

TIP Of the forms of data collection, the data provided by sensors is the easiest to make consistent. However, sensor data is often inconsistent because vendors keep adding functionality as a means of differentiation. The solution to this problem is better data standards so that vendors must adhere to certain specifics when creating data. Standards efforts are ongoing, but it pays to ensure that the sensors you use to collect data all rely on the same standards to ensure that you obtain consistent input.

## Using Reliable Sources

The word *reliable* seems so easy to define, yet so hard to implement. Something is reliable when the results it produces are both expected and consistent. A reliable data source produces mundane data that contains no surprises; no one is shocked in the least by the outcome. On the other hand, depending on your perspective, it could actually be a good thing that most people aren't yawning and then falling asleep when reviewing data. That's because the surprises make the data worth analyzing and reviewing. Consequently, data has an aspect of duality. You want reliable, mundane, fully anticipated data that simply confirms what you already know, but the unexpected is what makes collecting the data useful in the first place.

You can also define reliability by the number of failure points contained in any measured resource. More failure points automatically mean lower reliability if you have two data sources of equal reliability. Given that general data analysis, AI, machine learning, and deep learning all require huge amounts of information, the methodology used automatically reduces the reliability of such data because you have more failure points to consider. Consequently, you must have data from highly reliable sources of the correct type.

Scientists began fighting against impressive amounts of data for years before anyone coined the term *big data.* At that point, the Internet didn't produce the vast sums for data that it does today. Remember that big data is not just simply a fad created by software and hardware vendors but has a basis in many of the following fields:

- **Astronomy:** Consider the data received from spacecraft on a mission (such as *Voyager* or *Galileo*) and all the data received from radio telescopes, which are specialized antennas used to receive radio waves from astronomical bodies. A common example is the Search for Extraterrestrial Intelligence (SETI) project (`https://www.seti.org/`), which looks for extraterrestrial signals by observing radio frequencies arriving from space. The amount of data received and the computer power used to analyze a portion of the sky for a single hour is impressive (`http://www.setileague.org/askdr/howmuch.htm`). If aliens are out there, it's very hard to spot them. (The movie *Contact,* which you can read about at `https://www.amazon.com/exec/obidos/ASIN/B002GHHHKQ/datacservip0f-20/`, explores what could happen should humans actually intercept a signal.)
- **Meteorology:** Think about trying to predict weather for the near term given the large number of required measures, such as temperature, atmospheric pressure, humidity, winds, and precipitation at different times, locations, and altitudes. Weather forecasting is really one of the first problems in big data, and quite a relevant one. According to Weather Analytics, a company that provides climate data, more than 33 percent of the Worldwide Gross Domestic Product (GDP) is determined by how weather conditions affect agriculture, fishing, tourism, and transportation, just to name a few. Dating back to the 1950s, the first supercomputers of the time were used to crunch as much as data as possible because, in meteorology, the more data, the more accurate the forecast. That's the reason everyone is amassing more storage and processing capacity, as you can read in this story regarding the Korean Meteorological Association (`https://www.wired.com/insights/2013/02/how-big-data-can-boost-weather-forecasting/`) for weather forecasting and studying climate change.
- **Physics:** Consider the large amounts of data produced by experiments using particle accelerators in an attempt to determine the structure of matter, space,

and time. For example, the Large Hadron Collider (`https://home.cern/topics/large-hadron-collider`), the largest particle accelerator ever created, produces 15PB (petabytes) of data every year as a result of particle collisions (`https://home.cern/science/computing`).

- **Genomics:** Sequencing a single DNA strand, which means determining the precise order of the many combinations of the four bases — adenine, guanine, cytosine, and thymine — that constitute the structure of the associated molecule, requires quite a lot of data. For instance, a single chromosome, a structure containing the DNA in the cell, may require from 50MB to 300MB. A human being normally has 46 chromosomes, and the DNA data for just one person consumes an entire DVD. Just imagine the massive storage required to document the DNA data of a large number of people or to sequence other life forms on earth (`https://www.wired.com/2013/10/big-data-biology/`).
- **Oceanography:** Gathers data from the many sensors placed in the oceans to measure statistics, such as temperature and currents, using hydrophones and other sensors. This data even includes sounds for acoustic monitoring for scientific purposes (discovering characteristics about fish, whales, and plankton) and military defense purposes (finding sneaky submarines from other countries). You can have a sneak peek at this old surveillance problem, which is turning more complex and digital, by reading this article: `https://www.theatlantic.com/technology/archive/2014/08/listening-in-the-navy-is-tracking-ocean-sounds-collected-by-scientists/378630/`.
- **Satellites:** Recording images from the entire globe and sending them back to earth to monitor the Earth's surface and its atmosphere isn't a new business (TIROS 1, the first satellite to send back images and data, dates back to 1960). Over the years, however, the world has launched more than 1,400 active satellites that provide earth observation. The amount of data arriving on earth is astonishing and serves both military (surveillance) and civilian purposes, such as tracking economic development, monitoring agriculture, and monitoring changes and risks. A single European Space Agency's satellite, Sentinel 1A, generates 5PB of data during two years of operation, as you can read from `https://spaceflightnow.com/2016/04/28/europes-sentinel-satellites-generating-huge-big-data-archive/`).

All these data sources have one thing in common: Someone collects and stores the data as static information (once collected, the data doesn't change). This means that if errors are found, correcting them with an overall increase in reliability is possible. The next section of this chapter discusses dynamic data, which isn't nearly so easy to make reliable.

REMEMBER What you need to take away from this section is that you likely deal with immense amounts of data from various sources that could have any number of errors. Finding these errors in such huge quantities is nearly impossible. Using the most reliable sources that you can will increase the overall quality of the original data, reducing the effect of individual data failure points. In other words, sources that provide consistent data are more valuable than sources that don't.

## Verifying Dynamic Data Sources

Dynamic data provides you with timely sources of information to use for analysis. The trade-off is trying to manage all that data. When data flows in huge amounts, storing it all may be difficult or even impossible. In fact, storing it all might not even be useful. Here are some figures of just some of what you can expect to happen within a single minute on the Internet:

- 150 million emails sent
- 350,000 new tweets sent on Twitter
- 2.4 million queries requested on Google
- 700,000 people logged in to their account on Facebook

The following sections consider the effects of such huge amounts of dynamic data used for analysis purposes. Because dynamic data constantly changes, it presents significant management issues that you don't encounter with static data.

### Considering the problem

Dynamic data likely makes up more of the data you use for various forms of analysis today. When working with dynamic data, you often don't provide long-term storage because doing so would prove impossible and the data has a definite shelf life. In addition, the related data sources continue to churn out new data at an incredible rate. Consider the problems associated with trying to store data from these sources:

- As reported by the National Security Agency (NSA), the amount of information flowing through the Internet every day from all over the world amounted to

1,826PB of data in 2013, and 1.6 percent of it consisted of emails and telephone calls. To assure national security, the NSA must verify the content of at least 0.025 percent of all emails and phone calls (looking for key words that could signal something like a terrorist plot). That still amounts to 25PB per year, which equates to 37,500 CD-ROMs every year of data stored and analyzed (and that's growing). You can read the full story at `https://www.business-standard.com/article/news-ani/nsa-claims-analysts-look-at-only-0-00004-of-world-s-internet-traffic-for-surveillance-113081100`.

- The Internet of Things (IoT) is a reality. You may have heard the term many times in the past, but now the growth of items connected to the Internet is exploding. The idea is to put sensors and transmitters on everything and use the data to both better control what happens in the world and to make objects smarter. Transmitting devices are getting tinier, cheaper, and less power demanding; some are already so small that they can be put everywhere. (Just look at the ant-sized radio developed by Stanford engineers at `https://news.stanford.edu/news/2014/september/ant-radio-arbabian-090914.html`.) Experts estimate that by 2020, there will be six times as many connected things on earth as there are people, but many research companies and think tanks are already revisiting those figures.

Given such volumes of data, accumulating the data all day for incremental analysis might not seem efficient. In the past, a vendor would collect the information and store it in a data warehouse for batch analysis. However, useful data queries tend to ask about the most recent data in the stream, and data becomes less useful when it ages. (In some sectors, like financial, a day can be a lot of time.) In fact, many marketing firms analyze data on a constant basis to determine the effectiveness of marketing campaigns. Data collected just a few minutes ago might see use in tweaking a campaign to make it more effective (see the article at `https://www.sas.com/en_us/insights/articles/marketing/do-marketers-need-real-time-analytics.html` for details).

Moreover, you can expect even more data to arrive tomorrow (the amount of data increases daily) and that makes it difficult, if not impossible, to pull data from repositories as you push new data in. Pulling old data from repositories as fresh data pours in is akin to the punishment of Sisyphus. Sisyphus, as a Greek myth narrative, received a terrible punishment from the god Zeus: being forced to eternally roll an immense boulder up on the top of a hill, only to watch it roll back down each time (see `http://www.mythweb.com/encyc/entries/sisyphus.html` for additional details).

Sometimes, rendering things even more impossible to handle, data can arrive so fast and in such large quantities that writing it to disk is impossible: New information arrives faster than the time required to write it to the hard disk. This is a problem typical of particle experiments with particle accelerators such as the Large Hadron Collider, requiring scientists to decide what data to keep (`https://home.cern/science/computing/processing-what-record`). Of course, you may queue data for some time, but not for too long, because the queue will quickly grow and become impossible to maintain. For instance, if kept in memory, queue data will soon lead to an out-of-memory error.

REMEMBER Because new data flows may render older processing techniques used on static data obsolete, and procrastination is not a solution, people have devised multiple strategies to deal instantaneously with massive and changeable data amounts. People use three ways to deal with large amounts of data:

- **Stored:** Some data is stored because it may help answer unclear questions later. This method relies on techniques to store it immediately and analyze it later very fast, no matter how massive it is.
- **Summarized:** Some data is summarized because keeping it all as it is makes no sense; only the important data is kept.
- **Consumed:** The remaining data is consumed because its usage is predetermined. Algorithms can instantly read, digest, and turn the data into information. After that, the system forgets the data forever.

TIP When talking of massive data arriving into a computer system, you will often hear it compared to water: streaming data, data streams, data fire hose. You discover how working with a data stream is like consuming tap water: Opening the tap lets you store the water in cups or drinking bottles, or you can use it for cooking, scrubbing food, cleaning plates, or washing hands. In any case, most or all of the water is gone, yet it proves very useful and indeed vital.

## Analyzing streams with the right recipe

Streaming data needs streaming algorithms, and the key thing to know about streaming algorithms is that, apart from a few measures that it can compute exactly, a streaming algorithm necessarily provides approximate results. The algorithm output is almost correct, not quite guessing the correct answer, but close to it.

When dealing with streams, you clearly have to concentrate only on the measures of interest and leave out many details. You could be interested in a statistical measurement, such as mean, minimum, or maximum. Moreover, you could want to count elements in the stream or distinguish old information from new. There are many algorithms to use, depending on the problem, yet the recipes always use the same ingredients. The trick of cooking the perfect stream is to use one or all of these algorithmic tools as ingredients:

- **Sampling:** Reduce your stream to a more manageable data size; represent the entire stream or the most recent observations using a shifting data window.
- **Hashing:** Reduce infinite stream variety to a limited set of simple integer numbers.
- **Sketching:** Create a short summary of the measure you need, removing the less useful details. This approach lets you leverage a simple working storage, which can be your computer's main memory or its hard disk.

Another characteristic to keep in mind about algorithms operating on streams is their simplicity and low computational complexity. Data streams can be quite fast. Algorithms that require too many calculations can miss essential data, which means that the data is gone forever. When you view the situation in this light, you can appreciate how hash functions prove useful because they're prompt in transforming inputs into something easier to handle and search for both operations. You can also appreciate the sketching and sampling techniques, which bring about the idea of *lossy compression* that enables you to represent something complex by using a simpler form. You lose some detail but save a great deal of computer time and storage.

Sampling means drawing a limited set of examples from your stream and treating them as if they represented the entire stream. It is a well-known tool in statistics through which you can make inferences on a larger context (technically called the universe or the population) by using a small part of it.

## Looking for New Data Collection Trends

Because data is so valuable and users are sometimes adverse to giving it up, vendors constantly find new ways to collect data. One such method comes down to spying. Microsoft, for example, was recently accused (yet again) of spying on Windows 10 users even when the user doesn't want to share the data (see `https://www.extremetech.com/computing/282263-microsoft-windows-10-data-collection` for details). Lest you think that Microsoft is solely interested in your computing concerns, think again. The data it admits to collecting (and there is likely more) is pretty amazing (see `https://www.theverge.com/2017/4/5/15188636/microsoft-windows-10-data-collection-documents-privacy-concerns` for details).

Microsoft's data gathering doesn't stop with your Windows 10 actions; it also collects data with Cortana, the personal assistant (see `https://www.computerworld.com/article/3106863/cortana-the-spy-in-windows-10.html`). Mind you, Alexa is accused of doing the same thing (`https://www.washingtonpost.com/news/powerpost/paloma/the-technology-202/2019/05/06/the-technology-202-alexa-are-you-spying-on-me-here-s-why-smart-speakers-raise-serious-privacy-concerns/5ccf46a9a7a0a46cfe152c3c/`). Google, likewise, does the same thing (see `https://www.consumerwatchdog.org/privacy-technology/how-google-and-amazon-are-spying-you`). So, one of the trends the vendors are using is spying, and it doesn't stop with Microsoft, nor does it stop with the obvious spying sources.

It might actually be possible to write an entire book on the ways in which people are spying on you, but that would make for a very paranoid book, and there are other new data collection trends to consider. You may have noticed that you get more email from everyone about the services or products you were provided. Everyone wants you to provide free information about your experiences in one of these forms:

- **Close-ended surveys:** A close-ended survey is one in which the questions have specific answers that you check mark. The advantage is greater consistency of feedback. The disadvantage is that you can't learn anything beyond the predefined answers.

- **Open-ended surveys:** An open-ended survey is one in which the questions rely on text boxes in which the user enters data manually. In some cases, this form of survey enables you to find new information, but at the cost of consistency, reliability, and cleanliness of the data.
- **One-on-one interviews:** Someone calls you or approaches you at a place like the mall and talks to you. When the interviewer is well trained, you obtain consistent data and can also discover new information. However, the quality of this information comes at the cost of paying someone to obtain it.
- **Focus group:** Three or more people meet with an interviewer to discuss a topic (including products). Because the interviewer acts as a moderator, the consistency, reliability, and cleanliness of the data remain high and the costs are lower. However, now the data suffers contamination from the interaction between members of the focus group.
- **Direct observation:** No conversation occurs in this case; someone monitors the interactions of another party with a product or service and records the responses using a script. However, because you now rely on a third party to interpret someone else's actions, you have a problem with contamination in the form of bias. In addition, if the subject of the observation is aware of being monitored, the interactions likely won't reflect reality.

REMEMBER These are just a few of the methods that are seeing greater use in data collection today. They're just the tip of the iceberg. The concepts you should take away from this section is that no perfect means for collecting some types of data exists and that all data collection methods require some sort of participative event.

## Weeding Old Data

Contrary to fine wine, data doesn't age well and you need to think about methods of keeping it fresh. The problem is most apparent when working with static data. You collect data from a source like a robotic mission to Mars, and then use it to make decisions about future missions. The problem is determining when that data is too old. At some point, the data becomes useless because it no longer reflects state-of-the-art technology.

The problem with old data is monumental because you can't really create a rule that says the data is actually useless. You may not be able to use data for a current analysis for a line of business decision, but the same data may have historical value. In addition, old data often provides examples for use in creating models that reflect special, uncommon events. Consequently, the rule that states that you really don't need something until after you throw it out comes into play. Of course, you could always archive the data in some unused location for all eternity, but that's not really practical, either.

REMEMBER When thinking about old data, you must consider a number of factors that determine whether the data isn't useful any longer. These factors include (but aren't limited to):

- Technology to which the data refers
- Environment in which the data is collected
- Attitudes of society as a whole
- Use of data within an analysis
- Cost of storing the data versus value gained
- Legal or other requirements
- Data format with regard to analysis techniques

## Considering the Need for Randomness

Oddly enough, creating a consistent analysis often means training your algorithm using randomized data so that it doesn't learn a specific pattern. Randomization relies on the capability by your computer to generate random numbers, which means creating the number without a plan. Therefore, a random number is unpredictable, and as you generate subsequent random numbers, they shouldn't relate to each other.

REMEMBER However, randomness is hard to achieve. Even when you throw dice, the result can't be completely unexpected because of the way you hold the dice, the way you throw them, and the fact that the dice aren't perfectly shaped. Computers aren't good at creating random numbers, either. They generate randomness by using algorithms or pseudorandom tables (which work by using a *seed* value as a starting point, a number equivalent to an index) because a computer can't create a truly random number. Computers are deterministic machines; everything inside them responds to a well-defined response pattern, which means that it imitates randomness in some way.

### Considering why randomization is needed

Even if a computer can't create true randomness, streams of pseudorandom numbers (numbers that appear as random but that are somehow predetermined) can still make the difference in many computer science problems. Any algorithm that employs randomness in its logic can appear as a randomized algorithm, no matter whether randomness determines its results, improves performance, or mitigates the risk of failing by providing a solution in certain cases.

Usually you find randomness employed in selecting input data, the start point of the optimization, or the number and kind of operations to apply to the data. When randomness is a core part of the algorithm logic and not just an aid to its performance, the expected running time of the algorithm and even its results may become uncertain and subject to randomness, too; for instance, an algorithm may provide different, though equally good, results during each run. It's therefore useful to distinguish between kinds of randomized solutions, each one named after iconic gambling locations:

- **Las Vegas:** These algorithms are notable for using random inputs or resources to provide the correct problem answer every time. Obtaining a result may take an uncertain amount of time because of its random procedures. An example is the Quicksort algorithm.
- **Monte Carlo:** Because of their use of randomness, Monte Carlo algorithms may not provide a correct answer or even an answer at all, although these outcomes seldom happen. Because the result is uncertain, a maximum number of trials in their running time may bind them. Monte Carlo algorithms demonstrate that algorithms do not necessarily always successfully solve the problems they are supposed to. An example is the Solovay–Strassen primality test.
- **Atlantic City:** These algorithms provide a correct problem answer at least 75 percent of the time. Monte Carlo algorithms are always fast but not always correct, and Las Vegas algorithms are always correct but not always fast. People therefore think of Atlantic City algorithms as halfway between the two because they are usually both fast and correct. This class of algorithms was introduced in 1982 by J. Finn in an unpublished manuscript entitled *Comparison of*

*Probabilistic Test for Primality.* Created for theoretical reasons to test for prime numbers, this class comprises hard-to-design solutions, thus very few of them exist today.

### Understanding how probability works

Probability tells you the likelihood of an event, which you normally express as a number. In this book, and generally in the field of probabilistic studies, the probability of an event is measured in the range between 0 (no probability that an event will occur) and 1 (certainty that an event will occur). Intermediate values, such as 0.25 or 0.75, indicate that the event will happen with a certain frequency under conditions that should lead to that event (referred to as *trials*). Even if a numeric range from 0 to 1 doesn't seem intuitive at first, working with probability over time makes the reason for using such a range easier to understand. When an event occurs with probability 0.25, you know that out of 100 trials, the event will happen 0.25 * 100 = 25 times.

For instance, when the probability of your favorite sports team winning is 0.75, you can use the number to determine the chances of success when your team plays a game against another team. You can even get more specific information, such as the probability of winning a certain tournament (your team has a 0.65 probability of winning a match in this tournament) or conditioned by another event (when you aren't playing on your home court, the probability of winning for your team decreases to 0.60).

Probabilities can tell you a lot about an event, and they're helpful for algorithms, too. In a randomized algorithmic approach, you may wonder when to stop an algorithm because it should have reached a solution. It's good to know how long to look for a solution before giving up. Probabilities can help you determine how many iterations you may need.

REMEMBER You commonly hear about probabilities as percentages in sports and economics, telling you that an event occurs a certain number of times after 100 trials. It's exactly the same probability no matter whether you express it as 0.25 or 25 percent. That's just a matter of conventions. In gambling, you even hear about odds, which is another way of expressing probability, where you compare the likelihood of an event (for example, having a certain horse win the race) against not having the event happen at all. In this case, you express 0.25 as 25 against 75 or in any other way resulting in the same ratio.

You can multiply a probability for a number of trials and get an estimated number of occurrences of the event, but by doing the inverse, you can empirically estimate a probability. Perform a certain number of trials, observe each of them, and count the number of times an event occurs. The ratio between the number of occurrences and the number of trials is your probability estimate. For instance, the probability 0.25 is the probability of picking a certain suit when choosing a card randomly from a deck of cards. French playing cards (the most widely used type of deck, which also appears in America and Britain) provide a classic example for explaining probabilities. (The Italians, Germans, and Swiss, for example, use decks with different suits, which you can read about at `http://healthy.uwaterloo.ca/museum/VirtualExhibits/Playing%20Cards/decks/index.htm` The deck contains 52 cards equally distributed into four suits: clubs and spades, which are black, and diamonds and hearts, which are red. If you want to determine the probability of picking an ace, you must consider that, by picking cards from a deck, you will observe four aces, which affects the probability of picking an ace (rather than a specific ace, such as the ace of hearts). Your trials at picking the cards are 52 (the number of cards); therefore the answer in terms of probability is 4/52 = 0.077.

TIP You can get a more reliable estimate of an empirical probability by using a larger number of trials. When using a few trials, you may not get a correct estimate of the event probability because of the influence of chance. As the number of trials grows, event observations will get nearer to the true probability of the event itself. The principle there is a generating process behind events. To understand how the generating process works, you need many trials. Using trials in such a way is also known as *sampling* from a probabilistic distribution.